

Global Radial Monotonicity for Leading-Index-One Double Polylogarithms

Reflected Green curvature, Cauchy-mode flow,
odd-moment inequalities, and exact weight-generating identities

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Abstract

For every real $b > 0$, let $\theta_{1,b}(\rho)$ be the unique zero of $\text{Im Li}_{1,b}(\rho e^{i\theta}, 1)$ on the upper semicircle of radius $0 < \rho \leq 1$. We prove the complete leading-index-one part of the normalized-radius conjecture stated in the inspected ProveIt manuscript: $\eta_{1,b}(\rho) = \cos \theta_{1,b}(\rho)/\rho$ is strictly decreasing for $b > 1$, strictly increasing for $0 < b < 1$, and identically $1/2$ for $b = 1$. The proof is global in radius, not a Taylor-series sign argument.

A Dirichlet Green transform converts the polylogarithm into a second-order Stieltjes integral with a strictly concave positive density. In inverse coordinates, its imaginary-part zero is exactly the mode of a Cauchy-smoothed density. A reflected-curvature criterion proves strict motion of that mode, and a separate implicit derivative converts it to the required radial sign. The criterion applies more generally to Student-type kernels and to two monotone sectors of Hurwitz-shifted inner orders. It also proves the sign of every odd central moment of an explicit harmonic-moment probability law.

We obtain an exact upper-half-plane parametrization, an all-circle cutoff for $b \geq 1$, a two-term radial correction, and opposite radial-curvature signs at $b = 2$ and $b = 4$. An independent weight-generating identity yields closed sums over every odd and every even inner order, including sums at $z = -1$ and $z = 1/4$. The package contains self-contained proofs, 18 rational root certificates, 1,938 exact finite regression checks, five symbolic checks, numerical diagnostics, and an additive manuscript fragment. The arguments do not resolve the remaining higher-outer-order radial problem or the S6 and S8 arithmetic candidates.

Status. Research manuscript with ordinary mathematical proofs; not independently peer reviewed or proof-assistant formalized. The claimed advance is relative to the explicitly inspected source section. No claim of worldwide priority, exhaustive repository audit, or inspection of all incoming archive contents is made.

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1 The question and the precise advance

For $b > 0$, write

$$F_b(z) = F_{1,b}(z) = \text{Li}_{1,b}(z, 1) = \sum_{n=2}^{\infty} \frac{H_{n-1}^{(b)}}{n} z^n, \quad H_N^{(b)} = \sum_{k=1}^N k^{-b}. \quad (1.1)$$

Initially $|z| < 1$; throughout the paper the continuation is the one on $\mathbb{C} \setminus [1, \infty)$ that agrees with this series. The ordinary depth-one polylogarithm and logarithm use the conventions of DLMF 25.12 [8]. A “zero” of an angular imaginary part is not a zero of the complex function itself.

The ProveIt chapter fragment `chapters/05-real-local-radius.tex` states a normalized-radius conjecture and proves the sign of its first nonconstant local coefficient [1]. At leading index one it predicts exactly the three cases in the next theorem. The general signed-kernel chapter already establishes uniqueness and simplicity of the relevant angular zero [2]. We reprove the portion needed here to make all subsequent sign arguments self-contained.

Theorem 1.1 (Complete leading-index-one radial law). *For every $b > 0$ and $0 < \rho \leq 1$ there is exactly one $\theta_{1,b}(\rho) \in (0, \pi)$ such that*

$$\text{Im } F_b(\rho e^{i\theta_{1,b}(\rho)}) = 0.$$

Set $\eta_b(\rho) = \cos \theta_{1,b}(\rho)/\rho$. Then

$$\eta'_b(\rho) \begin{cases} < 0, & b > 1, \\ = 0, & b = 1, \\ > 0, & 0 < b < 1. \end{cases} \quad \eta_1(\rho) = \frac{1}{2}. \quad (1.2)$$

The function extends real analytically as an even function near zero, with

$$\eta_b(0) = \mu_b := \frac{1 + 2^{-b}}{3}. \quad (1.3)$$

Let r_b be the unique maximum point of the density Q_b in (2.9). For all positive disk radii,

$$0 < r_b < \eta_b(\rho) < \mu_b < \frac{1}{2}, \quad b > 1, \quad (1.4)$$

$$\frac{1}{2} < \mu_b < \eta_b(\rho) < r_b < 1, \quad 0 < b < 1. \quad (1.5)$$

At $\rho = 1$, the derivative means the derivative of the local analytic continuation of the same simple root; in particular the corresponding one-sided derivative has the displayed sign.

The proof occupies Sections 2–5. Its main new mechanism is an ordered sign change of a reflected density. Mere unimodality is not sufficient to infer the direction of motion of a mode; and motion in inverse height does not, by itself, imply motion in disk radius. These two additional steps are proved explicitly.

1.1 Additional results and boundaries of the claim

The same method proves a mode-flow theorem for a class of Green densities and every kernel $((x - u)^2 + y^2)^{-\alpha}$, $\alpha > 0$. It gives the whole imaginary-zero curve in the upper half-plane, including

a finite-radius cutoff when $b \geq 1$. A probability measure with raw moments $2H_{j+1}^{(b)}/((j+1)(j+2))$ has all nontrivial odd central moments of the sign of $b - 1$. The manuscript's local coefficient is exactly minus twice its third central moment. Nevertheless, the next radial coefficient changes sign: monotonicity does not entail a universal curvature sign.

A separate exact generating function sums over the inner order. Its half-integer specializations give

$$\sum_{j \geq 0} 4^{-j} F_{2j+1}(-1) = \frac{\pi}{2} - \frac{\pi^2}{8}, \quad (1.6)$$

$$\sum_{j \geq 0} 4^{-j} F_{2j+2}(-1) = \pi - 4 \log 2. \quad (1.7)$$

These are proved identities, not integer-relation guesses. Their functional versions hold on the principal slit plane.

Nothing here proves the complete normalized-radius assertion at larger outer orders. In particular, this paper does not promote the S6 or S8 period relations from numerically supported candidates to theorems. The source README expressly keeps those relations conjectural [3]. The distinction is important: the present advance is an analytic monotonicity theorem, not a finite-basis reduction of those constants.

1.2 Literature and source scope

Ibragimov's strong-unimodality theorem relates log-concavity to preservation of unimodality under convolution [6]. Our proof of unique smoothed modes uses that classical mechanism and supplies the strict second-derivative statement needed for differentiation. We do not claim strong unimodality itself as a new result. Hausdorff moment generating functions and their Pick-function structure provide related background in Liu–Pego [7]; the consulted arXiv version includes their 2025 corrigendum. Neither external theorem is used as an unproved step in the radial argument.

The source pin is the ProveIt commit

1539f353af88bdfd63292bb970ac8f585b15794d.

The local-radius fragment was re-read at that exact pin and has blob 30e3e31944be5bc0d03a8485386dd1608c075507. The manuscript README, relevant analytic fragments, and incoming intake README and directory listing were also inspected. The incoming ZIP members were not available for content-level inspection in this session. Accordingly, novelty is assessed against the named manuscript statements, not against an asserted exhaustive inventory of all incoming results.

2 A Green transform with harmonic moments

2.1 The general density and its endpoint control

Definition 2.1. An admissible weight is a continuous positive function $w : (0, 1) \rightarrow (0, \infty)$ with $W = \int_0^1 w(t) dt < \infty$. Define

$$Q_w(u) = (1 - u) \int_0^u \frac{w(t)}{1 - t} dt + u \int_u^1 \frac{w(t)}{t} dt, \quad 0 < u < 1. \quad (2.1)$$

Set $Q_w(0) = Q_w(1) = 0$, and extend it by zero outside $[0, 1]$ whenever it occurs in a convolution or reflection.

The factors outside the integrals in (2.1) are essential. The separate endpoint integrals can be infinite at $u = 0$ or $u = 1$, but their displayed products have well-defined limits. This point matters for every $0 < b < 1$ in the polylogarithm application.

Lemma 2.2 (Green regularity and exact moments). *For every admissible w , Q_w is continuous on $[0, 1]$, absolutely continuous there, strictly positive on $(0, 1)$, and strictly concave. It is twice continuously differentiable in the open interval and*

$$Q'_w(u) = \int_u^1 \frac{w(t)}{t} dt - \int_0^u \frac{w(t)}{1-t} dt, \quad (2.2)$$

$$Q''_w(u) = -\frac{w(u)}{u(1-u)} < 0. \quad (2.3)$$

Its logarithm is strictly concave on $(0, 1)$. Moreover,

$$\int_0^1 u^j Q_w(u) du = \frac{1}{(j+1)(j+2)} \int_0^1 w(t) \frac{1-t^{j+1}}{1-t} dt, \quad j \geq 0. \quad (2.4)$$

In particular its mass is $W/2$.

Proof. Write $G(u, t) = \min(u, t)(1 - \max(u, t))$. Then

$$Q_w(u) = \int_0^1 \frac{G(u, t)}{t(1-t)} w(t) dt.$$

The ratio in this integrand is between zero and one, and tends to zero as u tends to either endpoint, for each fixed $t \in (0, 1)$. Dominated convergence proves continuity and the two zero limits, and also gives $Q_w \leq W$. Positivity is immediate.

Differentiating in the open interval gives (2.2) and (2.3). The two positive terms in (2.2) each have integral W over $u \in (0, 1)$, by Tonelli. Thus $Q'_w \in L^1(0, 1)$, and the interior fundamental theorem of calculus extends to both endpoints. Strict concavity follows from (2.3); strict log-concavity follows from

$$(\log Q_w)'' = \frac{Q''_w}{Q_w} - \left(\frac{Q'_w}{Q_w} \right)^2 < 0.$$

Finally, a direct integration of the two linear pieces of G gives

$$\int_0^1 u^j G(u, t) du = \frac{t - t^{j+2}}{(j+1)(j+2)}.$$

Tonelli now proves (2.4). This derivation does not assume that either unweighted endpoint integral in (2.2) is finite. $\square$

2.2 A signed measure and a second-order Stieltjes integral

Put $\kappa_w = -Q'_w$. It is integrable, strictly increasing, and has zero integral. It therefore has a unique strict negative-to-positive crossing $r_w \in (0, 1)$, which is the unique maximum point of Q_w . Define

$$\mathcal{F}_w(z) = z \int_0^1 \frac{\kappa_w(u)}{1-zu} du = z^2 \int_0^1 \frac{Q_w(u)}{(1-zu)^2} du. \quad (2.5)$$

Integration by parts proves the equality; absolute continuity and $Q_w(0) = Q_w(1) = 0$ remove the boundary terms. Both integrals are holomorphic on $\mathbb{C} \setminus [1, \infty)$. Indeed, on a compact subset of this domain, $|1 - zu|$ has a uniform positive lower bound for $0 \leq u \leq 1$.

The moments in Lemma 2.2 give the disk series

$$\mathcal{F}_w(z) = \sum_{n \geq 2} \frac{z^n}{n} \int_0^1 w(t) \frac{1 - t^{n-1}}{1 - t} dt. \quad (2.6)$$

They also show, directly, that

$$\int_0^1 u^{n-1} \kappa_w(u) du = (n-1) \int_0^1 u^{n-2} Q_w(u) du$$

for $n \geq 2$, while the $n = 1$ moment is zero.

Remark 2.3 (A positive first-order transform after a factor). The one-crossing property also gives

$$\mathcal{F}_w(z) = \frac{z^2}{1 - r_w z} \int_0^1 \frac{(u - r_w) \kappa_w(u)}{1 - zu} du. \quad (2.7)$$

The new numerator is positive except at its crossing. The integral is positive for real $z < 1$ and has the sign of $\text{Im } z$ in its imaginary part off the real axis. Thus $\mathcal{F}_w$ has no nonzero zeros in the slit plane. This is a useful direct check that the curves studied below concern imaginary parts, not complex zeros. Comparable nonvanishing results already occur in the source manuscript; the new contribution is their radial geometry.

2.3 Specialization to inner polylogarithm order

Take

$$w_b(u) = \frac{(-\log u)^{b-1}}{\Gamma(b)}, \quad b > 0. \quad (2.8)$$

The gamma integral gives

$$\int_0^1 u^k w_b(u) du = (k+1)^{-b}, \quad k \geq 0.$$

Consequently $W = 1$, (2.6) becomes (1.1), and the associated Green density has the explicit form

$$Q_b(u) = \frac{u(-\log u)^b}{\Gamma(b+1)} + \frac{1-u}{\Gamma(b)} \int_0^u \frac{(-\log t)^{b-1}}{1-t} dt. \quad (2.9)$$

The probability density $2Q_b$ therefore has moments

$$\mathbb{E}U_b^j = \frac{2H_{j+1}^{(b)}}{(j+1)(j+2)}, \quad j \geq 0, \quad (2.10)$$

and mean μ_b from (1.3). Its curvature is

$$-u(1-u)Q_b''(u) = w_b(u). \quad (2.11)$$

The right side is strictly decreasing when $b > 1$, strictly increasing when $0 < b < 1$, and constant when $b = 1$. This elementary observation will control a global zero curve.

For $b = 1$,

$$Q_1(u) = -u \log u - (1 - u) \log(1 - u), \quad F_1(z) = \frac{1}{2} \log^2(1 - z). \quad (2.12)$$

For a positive integer m , the density also has a finite polylogarithmic expression. With $T = -\log u$,

$$Q_m(u) = \frac{uT^m}{m!} + (1 - u) \sum_{j=0}^{m-1} \frac{T^j}{j!} \text{Li}_{m-j}(u). \quad (2.13)$$

To prove this, substitute $t = e^{-v}$ in the integral of (2.9), expand $(e^v - 1)^{-1}$ as a positive exponential series, and integrate $v^{m-1}e^{-kv}$ from T to infinity. Its finite incomplete-gamma formula gives exactly the displayed sum.

3 Unique modes under Cauchy and Student-type smoothing

For $\alpha > 0$, $y > 0$, define

$$A_{\alpha,y}(x) = \int_0^1 \frac{Q_w(u)}{((x-u)^2 + y^2)^\alpha} du. \quad (3.1)$$

The kernel is integrable on the whole line only when $\alpha > 1/2$, but (3.1) is finite and analytic for every $\alpha > 0$ because Q_w has compact support. Thus “Student-type” here describes the kernel shape, not a claim of probability normalization for every α .

Lemma 3.1 (Strict unique mode). *For every admissible w , every $\alpha > 0$, and every $y > 0$, $A_{\alpha,y}$ has exactly one stationary point $m_\alpha(y) \in (0, 1)$. It is a strict maximum and*

$$\partial_x^2 A_{\alpha,y}(m_\alpha(y)) < 0. \quad (3.2)$$

The function m_α is real analytic on $(0, \infty)$.

Proof. Let $K(v) = (v^2 + y^2)^{-\alpha}$. Then $K'(v)$ is positive for $v < 0$ and negative for $v > 0$. Therefore $A_x(x) > 0$ for $x \leq 0$ and $A_x(x) < 0$ for $x \geq 1$, so at least one stationary point lies between.

Fix any such point x . After $v = x - u$, the stationary equation is

$$\int_{x-1}^x Q_w(x-v) K'(v) dv = 0.$$

The signed integrand is positive on $v < 0$ and negative on $v > 0$. The logarithmic score

$$S(v) = \frac{Q'_w(x-v)}{Q_w(x-v)}$$

is strictly increasing, since $(\log Q_w)'' < 0$. Using the zero integral and subtracting $S(0)$ gives

$$\begin{aligned} A_{xx}(x) &= \int_{x-1}^x Q'_w(x-v) K'(v) dv \\ &= \int_{x-1}^x Q_w(x-v) K'(v) (S(v) - S(0)) dv < 0. \end{aligned}$$

The integration by parts implicit in the first equality is legitimate: Q_w vanishes at the endpoints, Q'_w is integrable, and K' is bounded on the finite integration range. The score need not remain bounded near the endpoints; the product is still integrable by this same L^1 bound.

Thus every zero of A_x crosses strictly from positive to negative. There cannot be two such zeros, because between two downward crossings a continuous derivative must return from negative to positive, forcing a zero that is not a strict downward crossing. This proves uniqueness and (3.2). The analytic implicit-function theorem applies to $A_x = 0$, since the integral is locally real analytic in (x, y) for $y > 0$. $\square$

Remark 3.2 (Which function is log-concave). It is Q_w , not the Cauchy kernel, whose log-concavity is used. The Cauchy kernel is not log-concave on the entire real line. Lemma 3.1 is a strict, directly proved version of the strong-unimodality mechanism associated with Ibragimov [6]. It does not yet determine a direction for $m'_\alpha(y)$.

4 Reflected curvature and strict mode flow

The central step is a geometric consequence of the normalized curvature $-u(1-u)Q''_w(u) = w(u)$.

Lemma 4.1 (A reflected single crossing). *Suppose w is nonconstant and nonincreasing. For $0 < x < 1/2$, put*

$$\Delta_x(v) = Q_w(x+v) - Q_w(x-v), \quad v \geq 0, \quad (4.1)$$

using the zero extension. Then Δ_x is strictly convex on $(0, x)$, $\Delta_x(0) = 0$, and $\Delta_x(x) = Q_w(2x) > 0$. It is positive for $x \leq v < 1-x$ and zero for $v \geq 1-x$. If it takes a negative value, it has exactly one negative-to-positive crossing $v_0 \in (0, x)$ on its nonzero support. At $x = 1/2$, $\Delta_{1/2}(v) < 0$ for $0 < v < 1/2$.

Proof. For $0 < v < x < 1/2$, (2.3) gives

$$\Delta''_x(v) = \frac{w(x-v)}{(x-v)(1-x+v)} - \frac{w(x+v)}{(x+v)(1-x-v)} > 0. \quad (4.2)$$

Indeed, the numerator in the first term is no smaller, whereas its denominator is strictly smaller, because the difference of the two denominators is $2v(1-2x) > 0$. The support and endpoint statements follow directly from the zero extension.

Strict convexity and $\Delta_x(0) = 0$ imply that $\Delta_x(v)/v$ is strictly increasing on $(0, x)$. Thus a negative value and the positive value at x force one and only one positive crossing before x . Beyond x the difference is already positive.

At $x = 1/2$, the denominator difference vanishes and the numerator ordering gives $\Delta'' \geq 0$. It is not identically zero: otherwise $w(1/2-v) = w(1/2+v)$ for all v , and a nonincreasing symmetric function would be constant. Since both endpoints of Δ are zero, convexity and nonlinearity make Δ strictly negative in the open interval. $\square$

Theorem 4.2 (Monotone Green curvature forces monotone mode flow). *Let w be admissible and $\alpha > 0$. If w is nonconstant and nonincreasing, then*

$$0 < m_\alpha(y) < \frac{1}{2}, \quad m'_\alpha(y) > 0 \quad (y > 0). \quad (4.3)$$

If w is nonconstant and nondecreasing, then

$$\frac{1}{2} < m_\alpha(y) < 1, \quad m'_\alpha(y) < 0 \quad (y > 0). \quad (4.4)$$

If w is constant, $m_\alpha(y) = 1/2$ for all y .

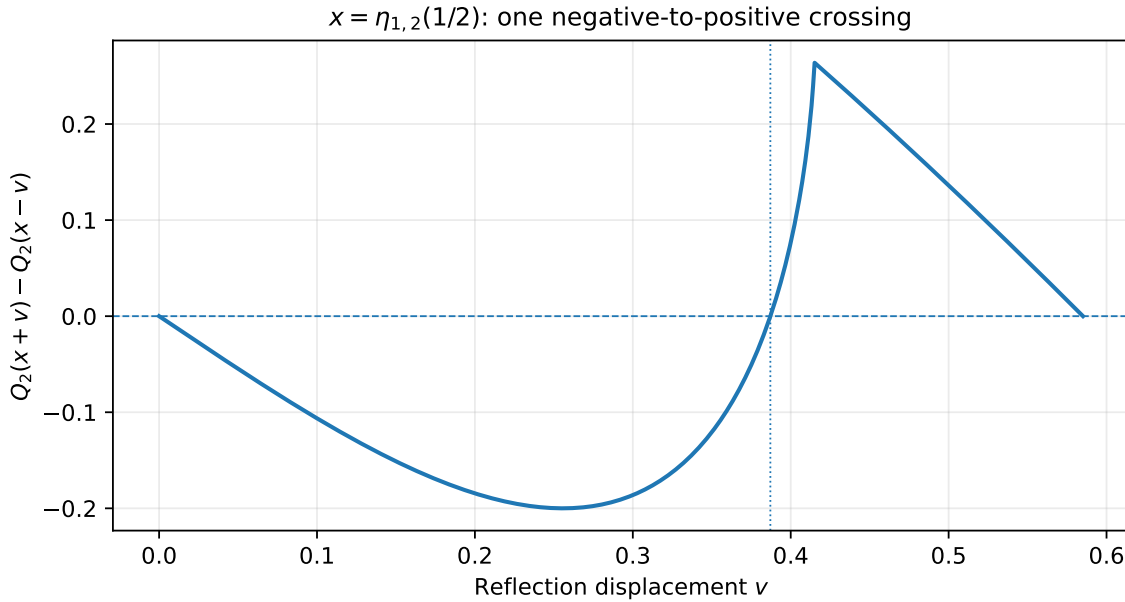


Figure 1: The decisive sign change, illustrated with $b = 2$ and $x = \eta_2(1/2)$. The graph is a floating-point illustration, not a certificate. The unique negative-to-positive crossing is proved by (4.2); the positive derivative in (4.6) follows by weighting this same crossing.

Proof. First suppose w is nonconstant and nonincreasing. Write $t = y^2$. Pairing the two sides of x in the derivative of (3.1) gives

$$A_x(x) = 2\alpha I_\alpha(x, t), \quad I_\alpha(x, t) = \int_0^\infty \frac{v \Delta_x(v)}{(v^2 + t)^{\alpha+1}} dv. \quad (4.5)$$

Lemma 4.1 gives $A_x(1/2) < 0$. By unique modehood, the mode $x = m_\alpha(y)$ lies below $1/2$.

At this mode, $I_\alpha = 0$. The positive part of Δ_x near and beyond $v = x$ is nonempty, so the zero integral forces a negative part. Lemma 4.1 therefore gives a unique crossing v_0 , with negative sign before and positive sign after it. Differentiate with respect to t at fixed x and subtract a constant multiple of the zero integral:

$$\partial_t I_\alpha(x, t) = \int_0^\infty \frac{v \Delta_x(v)}{(v^2 + t)^{\alpha+1}} \left(-\frac{\alpha+1}{v^2 + t} + \frac{\alpha+1}{v_0^2 + t} \right) dv > 0. \quad (4.6)$$

The parenthesized factor is strictly increasing in $v > 0$ and vanishes at v_0 ; hence its sign agrees with that of Δ_x . All differentiations are dominated because $t > 0$ and the support is bounded.

By Lemma 3.1, $I_{\alpha,x} < 0$ at the root. Implicit differentiation gives

$$\frac{dm_\alpha}{dt} = -\frac{I_{\alpha,t}}{I_{\alpha,x}} > 0, \quad m'_\alpha(y) = 2y \frac{dm_\alpha}{dt} > 0.$$

For a nondecreasing weight, reflect $u \mapsto 1 - u$. Its Green density is reflected as well, so every inequality reverses. A constant weight has a symmetric density, whose unique mode is the center. $\square$

4.1 The Cauchy mode interpolates between mode and mean

For $\alpha = 1$, multiplication by y/π gives the usual Poisson, or Cauchy, kernel

$$P_y(v) = \frac{y}{\pi(v^2 + y^2)}.$$

Write $m(y) = m_1(y)$ and let μ_w be the mean of the probability density $2Q_w/W$. Formula (2.4) gives

$$\mu_w = \frac{1}{3} \left(1 + \frac{\int_0^1 tw(t) dt}{W} \right). \quad (4.7)$$

Corollary 4.3 (Strict interpolation). *For every admissible w ,*

$$\lim_{y \downarrow 0} m(y) = r_w, \quad \lim_{y \rightarrow \infty} m(y) = \mu_w. \quad (4.8)$$

If w is nonconstant and nonincreasing, then

$$0 < r_w < m(y) < \mu_w < \frac{1}{2}.$$

If it is nonconstant and nondecreasing, then

$$\frac{1}{2} < \mu_w < m(y) < r_w < 1.$$

Proof. The zero extension of Q_w is continuous and compactly supported. Thus $P_y * Q_w \rightarrow Q_w$ uniformly as $y \downarrow 0$, by the elementary approximate-identity argument. Each maximum lies in $[0, 1]$, and any limit point of the maxima must maximize Q_w . Its strict concavity leaves only r_w .

For large y , the stationary equation is

$$0 = \int_0^1 (u - m(y)) Q_w(u) \left(1 + \frac{(u - m(y))^2}{y^2} \right)^{-2} du.$$

The factor in parentheses tends uniformly to one for $m(y) \in [0, 1]$. It follows that $m(y) \rightarrow \mu_w$.

Strict inequalities between the endpoint limits follow from Theorem 4.2. Finally, if w is nonconstant and nonincreasing, pairing t and $1 - t$ shows $\int_0^1 (t - 1/2)w(t) dt < 0$, so (4.7) gives $\mu_w < 1/2$. Reflection handles the other case. $\square$

5 From mode flow to the global radial theorem

5.1 The known angular root, with its derivative sign

Lemma 5.1 (Angular existence, uniqueness, and simplicity). *For every admissible w and $0 < \rho \leq 1$, the function $\text{Im } \mathcal{F}_w(\rho e^{i\theta})$ has exactly one zero on $0 < \theta < \pi$. With $c = \cos \theta$, this root lies in $0 < c < \rho$. It is simple, and the imaginary part has a negative angular derivative there.*

Proof. Let $\kappa = \kappa_w$, and put

$$J_\rho(c) = \int_0^1 \frac{\kappa(u)}{1 - 2\rho cu + \rho^2 u^2} du. \quad (5.1)$$

Then the imaginary part is $\rho \sin \theta J_\rho(c)$. For any strictly increasing function h , subtraction of $h(r_w)$ shows $\int h\kappa > 0$; for a strictly decreasing h the sign is negative. In particular, $J_\rho(0) < 0$. At $c = \rho$

the denominator's derivative in u is $-2\rho^2(1-u) < 0$, so its reciprocal is strictly increasing and $J_\rho(\rho) > 0$. If $\rho = 1$, this is interpreted as a positive, possibly infinite, endpoint limit. The negative part of κ is away from the singular endpoint; the positive part is treated by monotone convergence. No difference of divergent integrals is used.

Set $K_c(u) = (1 - 2\rho c u + \rho^2 u^2)^{-1}$. For $c' > c$,

$$\frac{d}{du} \frac{K_{c'}(u)}{K_c(u)} = \frac{2\rho(c' - c)(1 - \rho^2 u^2)}{(1 - 2\rho c' u + \rho^2 u^2)^2} > 0. \quad (5.2)$$

At a zero $J_\rho(c) = 0$, the signed measure $K_c \kappa du$ has zero mass and the same one-crossing sign pattern. The increasing ratio in (5.2) makes $J_\rho(c') > 0$ for all larger c' ; for smaller c' the sign is negative. This proves uniqueness. The score $2\rho u/(1 - 2\rho c u + \rho^2 u^2)$ is strictly increasing in u as well, so $J'_\rho(c) > 0$ at its zero. Multiplication by $dc/d\theta = -\sin \theta$ gives the negative angular derivative. $\square$

5.2 An exact inverse-coordinate identity

For z in the upper half-plane, write uniquely

$$z = \frac{1}{x - iy}, \quad x \in \mathbb{R}, \quad y > 0. \quad (5.3)$$

Formula (2.5) becomes $\mathcal{F}_w(z) = \int_0^1 Q_w(u)(x - u - iy)^{-2} du$. Taking its imaginary part gives the exact identity

$$\operatorname{Im} \mathcal{F}_w\left(\frac{1}{x - iy}\right) = -\pi \partial_x (P_y * Q_w)(x). \quad (5.4)$$

Thus every upper-half-plane imaginary-part zero, and no other one, has inverse coordinates $(m(y), y)$. The general upper-half-plane curve is already determined; the remaining task is to intersect it with circles centered at zero.

Theorem 5.2 (Global disk radial law for monotone Green curvature). *Under the assumptions of Theorem 4.2, let $\eta_w(\rho) = \cos \theta_w(\rho)/\rho$ be the root from Lemma 5.1. On $0 < \rho \leq 1$, it is strictly decreasing when w is nonconstant and nonincreasing, and strictly increasing when w is nonconstant and nondecreasing. For constant w it is identically $1/2$.*

Proof. Put $s = \rho^2$ and

$$x = \eta_w(\sqrt{s}), \quad t = y^2 = s^{-1} - x^2 > 0. \quad (5.5)$$

For the Cauchy case abbreviate

$$I(x, t) = \int_0^1 \frac{(u - x)Q_w(u)}{((u - x)^2 + t)^2} du, \quad \mathcal{I}(x, s) = I(x, s^{-1} - x^2).$$

At the zero, $I = 0$. The sign calculation (4.6) gives $I_t > 0$ for nonincreasing w and $I_t < 0$ for nondecreasing w .

The crucial point is the derivative in the disk coordinate, not merely I_x at fixed height. Since $\operatorname{Im} z = sy$, comparison of (5.1) with (5.4) yields

$$J_{\sqrt{s}}(\sqrt{s}x) = -\frac{2}{s} \mathcal{I}(x, s). \quad (5.6)$$

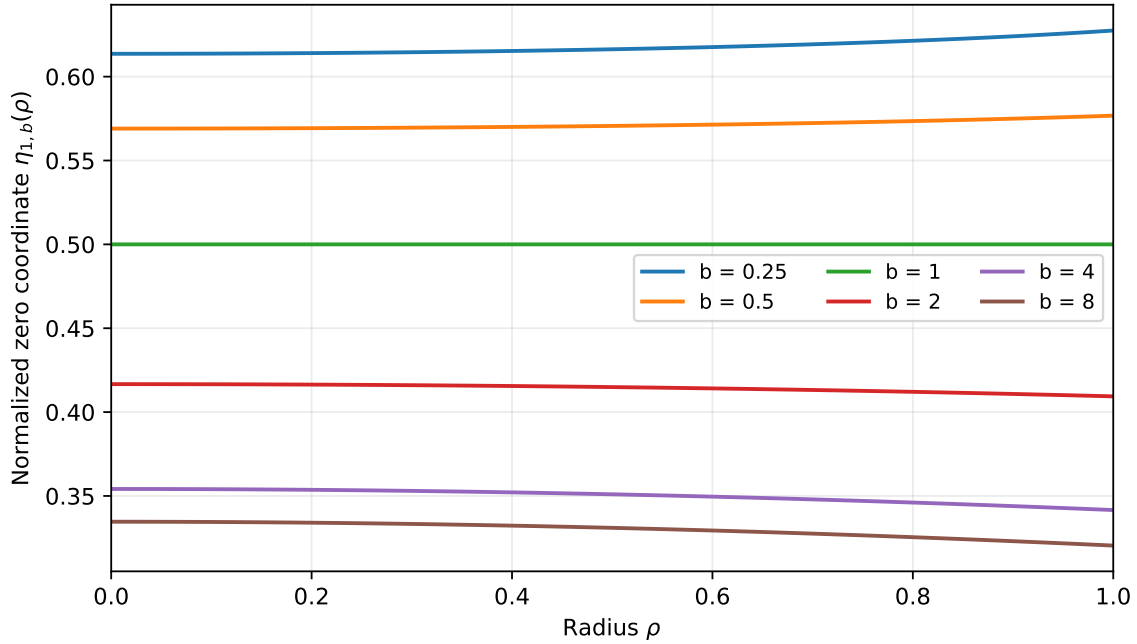


Figure 2: The normalized-radius law at six inner orders. All strict signs on the whole interval are proved in Theorem 1.1. The plotted values use 256-node generalized Gauss–Laguerre quadrature and are illustrative. Eighteen selected roots, including fractional inner orders, have separate rational certificates.

Lemma 5.1 therefore gives $\mathcal{I}_x < 0$ at the root. At fixed x , $\mathcal{I}_s = -s^{-2}I_t$, so implicit differentiation gives

$$\frac{d}{ds}\eta_w(\sqrt{s}) = \frac{s^{-2}I_t}{\mathcal{I}_x}. \quad (5.7)$$

This has precisely the asserted sign. Since $ds/d\rho = 2\rho > 0$, the sign in radius is the same. If w is constant, reflection symmetry makes the unique inverse mode $1/2$, hence the unique disk root has $\eta_w = 1/2$. $\square$

Proof of Theorem 1.1. Apply Theorem 5.2 to w_b in (2.8); its monotonicity is given by (2.11). The bounds follow from Corollary 4.3, because the root in (5.5) lies on the Cauchy-mode graph.

For the analytic extension at zero, write $z = s\eta + i\sqrt{s - s^2}\eta^2$. The polynomials $P_n = \text{Im}(z^n)/\text{Im } z$ satisfy $P_0 = 0$, $P_1 = 1$, and $P_n = 2s\eta P_{n-1} - sP_{n-2}$. Substituting the convergent disk series into J shows that J/s is analytic in (η, s) near $(\mu_b, 0)$ and starts with

$$\frac{J}{s} = \eta - \mu_b + O(s).$$

Here the coefficients of z^2, z^3 in F_b are $1/2$ and μ_b . The analytic implicit-function theorem proves the extension in s , and therefore the even extension in ρ . Simplicity away from the cut gives analyticity through every positive radius under discussion, including $\rho = 1$. $\square$

5.3 The whole upper-half-plane curve and the outer cutoff

Corollary 5.3 (A global parametrization and a sharp circle cutoff). *For every $b > 0$, the entire upper-half-plane zero set of $\text{Im } F_b$ is*

$$z_b(y) = \frac{1}{m_b(y) - iy}, \quad y > 0, \quad (5.8)$$

where m_b is the unique Cauchy mode associated with Q_b . It is a simple analytic curve. If $b > 1$, every circle of radius $0 < \rho < 1/r_b$ has exactly one such upper-half-plane zero, and circles of radius $\rho \geq 1/r_b$ have none. For $b = 1$ the cutoff is 2. For $b > 1$ the normalized coordinate is strictly decreasing throughout $0 < \rho < 1/r_b$, not only in the unit disk.

Proof. The parametrization follows from (5.4) and Lemma 3.1; it is injective because the inverse coordinate y is unique. Its derivative never vanishes: the derivative of $m_b(y) - iy$ has imaginary part -1 . For $b > 1$, $m_b > 0$ and $m'_b > 0$, so

$$R_b(y) = |z_b(y)| = (m_b(y)^2 + y^2)^{-1/2}$$

is strictly decreasing. Its limits are $1/r_b$ at $y \downarrow 0$ and zero at infinity. This proves the circle count and cutoff. Along the curve, $\eta_b(R_b(y)) = m_b(y)$, giving the strict radial sign on the whole interval. At $b = 1$, $m_b = 1/2$ and the same argument has cutoff 2. The limiting point $1/r_b$ lies on the excluded real cut, not in the upper half-plane; it is not an omitted upper-arc zero. $\square$

For $0 < b < 1$, the inverse coordinate $m_b(y)$ decreases, but this alone does not determine the sign of $R'_b(y)$ outside the disk. The complete inverse graph remains proved; a global circle-count assertion in this range is deliberately not made.

6 All odd central moments have a prescribed sign

The reflection argument also gives a hierarchy of exact inequalities, without introducing a smoothing kernel.

Theorem 6.1 (Reflected skew inequalities). *Let U have density $2Q_w/W$ and mean μ_w . Suppose w is nonconstant and nonincreasing. For every continuous strictly increasing function φ on $[0, 1]$,*

$$\mathbb{E}[(U - \mu_w)\varphi((U - \mu_w)^2)] > 0. \quad (6.1)$$

For a nonconstant nondecreasing weight the sign is negative. For a constant weight it is zero. In particular, for every integer $m \geq 1$ the central moment $\mathbb{E}(U - \mu_w)^{2m+1}$ has the same sign.

Proof. Consider the nonincreasing case. We know $\mu_w < 1/2$. The mean identity is

$$\int_0^\infty v \Delta_{\mu_w}(v) dv = \int_0^1 (u - \mu_w) Q_w(u) du = 0.$$

Since Δ_{μ_w} has a positive part, this forces a negative part. Lemma 4.1 supplies a unique crossing v_0 . Consequently

$$\int_0^\infty v \Delta_{\mu_w}(v) (\varphi(v^2) - \varphi(v_0^2)) dv > 0.$$

Division by $W/2$ proves (6.1). Choosing $\varphi(t) = t^m$ proves the moment assertion. Reflection handles the opposite monotonicity, and symmetry handles a constant weight. $\square$

Corollary 6.2 (Explicit harmonic inequalities). *For $b > 0$ and every integer $m \geq 1$, put $\mu_b = (1 + 2^{-b})/3$. Then*

$$2 \sum_{j=0}^{2m+1} \binom{2m+1}{j} (-\mu_b)^{2m+1-j} \frac{H_{j+1}^{(b)}}{(j+1)(j+2)} \quad (6.2)$$

is positive for $b > 1$, negative for $0 < b < 1$, and zero for $b = 1$.

Proof. Expression (6.2) is the binomial expansion of $\mathbb{E}(U_b - \mu_b)^{2m+1}$, using (2.10). Apply Theorem 6.1 to w_b . $\square$

The statement is a continuum theorem in b and an all-order theorem in m . Exact finite tests in the package are useful regression checks, but are not its proof.

Corollary 6.3 (An exponential version). *For every $\lambda > 0$,*

$$\mathbb{E}e^{\lambda(U_b - \mu_b)} - \mathbb{E}e^{-\lambda(U_b - \mu_b)}$$

has the sign of $b - 1$, with strictness for $b \neq 1$.

Proof. The function $\varphi(t) = \sinh(\lambda\sqrt{t})/\sqrt{t}$, defined at zero by continuity, is strictly increasing on $[0, 1]$: its power series has a positive constant and strictly positive higher coefficients. Equation (6.1) gives the result, since $(U - \mu)\varphi((U - \mu)^2) = \sinh(\lambda(U - \mu))$. $\square$

7 Radial expansions and a genuine curvature limitation

Write $M_j = \mathbb{E}(U - \mu)^j$ for the normalized Green law. These moments are finite because the support is compact.

Proposition 7.1 (Large-height mode expansion). *For $\alpha > 0$, put $p = \alpha + 1$. As $y \rightarrow \infty$,*

$$m_\alpha(y) = \mu - pM_3y^{-2} + \left(\frac{p(p+1)}{2}M_5 - 3p^2M_2M_3 \right) y^{-4} + O(y^{-6}). \quad (7.1)$$

No monotonicity assumption on w is required for this expansion.

Proof. Let $V = U - \mu$, $d = m_\alpha(y) - \mu$, and $h = y^{-2}$. The stationary equation is

$$0 = \mathbb{E}[(V - d)(1 + h(V - d)^2)^{-p}].$$

At $h = 0$ it has the simple solution $d = 0$, with derivative -1 in d . The analytic implicit-function theorem applies near $(0, 0)$; compact support makes the binomial expansion uniform there. Expanding through h^2 , using $d = O(h)$, gives

$$0 = -d - pM_3h + 3pM_2dh + \frac{p(p+1)}{2}M_5h^2 + O(h^3).$$

Solving successively for the coefficients of h and h^2 yields (7.1). $\square$

Corollary 7.2 (Radial expansion through fourth order). *For the leading-index-one polylogarithm, with moments of $2Q_b$,*

$$\eta_b(\rho) = \mu_b - 2M_3\rho^2 + [3M_5 - (12M_2 + 2\mu_b^2)M_3]\rho^4 + O_b(\rho^6). \quad (7.2)$$

Let $U = H_2^{(b)}$, $V = H_3^{(b)}$, $W = H_4^{(b)}$. Then

$$-2M_3 = \frac{UV}{3} - \frac{W}{5} - \frac{4U^3}{27} = K(1, b), \quad (7.3)$$

where K is the coefficient already recorded in the manuscript.

Proof. Take $\alpha = 1$, so $p = 2$, in (7.1), and use $y^{-2} = \rho^2/(1 - \rho^2\eta_b(\rho)^2) = \rho^2 + \mu_b^2\rho^4 + O(\rho^6)$. The raw moments give

$$M_3 = \frac{W}{10} - \frac{UV}{6} + \frac{2U^3}{27},$$

which proves (7.3). $\square$

This proves the $a = 1$ local sign by a conceptual positive-measure argument and strengthens it globally. It is not a correction to the existing formula: the existing formula agrees exactly.

Proposition 7.3 (No universal radial-curvature sign). *Let $s = \rho^2$. Although $\eta_b(\sqrt{s})$ decreases for every $b > 1$, it is strictly concave near $s = 0$ when $b = 2$ and strictly convex there when $b = 4$. More precisely,*

$$\eta_2(\rho) = \frac{5}{12} - \frac{1}{144}\rho^2 - \frac{263}{777600}\rho^4 + O(\rho^6), \quad (7.4)$$

$$\eta_4(\rho) = \frac{17}{48} - \frac{15893}{1244160}\rho^2 + \frac{504047}{3149280000}\rho^4 + O(\rho^6). \quad (7.5)$$

In particular $s \mapsto \eta_2(\sqrt{s})$ is not completely monotone on any interval $(0, \varepsilon)$.

Proof. Substitute the rational moments from (2.10) into (7.2). The two fourth-order coefficients are the exact fractions displayed. The second s derivative tends to twice that coefficient, giving the strict signs near zero. Complete monotonicity would require the second derivative to be nonnegative; the $b = 2$ expansion excludes it. $\square$

Thus a proposed upgrade from monotonicity to one fixed convexity or complete-monotonicity class would be false. No such upgrade is alleged to have been claimed by the inspected manuscript. The proposition is a useful boundary for future conjecture formation.

8 Hurwitz-shifted inner orders

For real $\lambda > -1$, $b > 0$, define

$$F_b^{(\lambda)}(z) = \sum_{n \geq 2} \frac{z^n}{n} \sum_{k=1}^{n-1} (k + \lambda)^{-b}. \quad (8.1)$$

The corresponding admissible weight is

$$w_{\lambda, b}(u) = \frac{u^\lambda (-\log u)^{b-1}}{\Gamma(b)}. \quad (8.2)$$

Its mass is $(1 + \lambda)^{-b}$; its Green transform and slit-plane continuation are defined as before. The logarithmic derivative is

$$\frac{d}{du} \log w_{\lambda,b}(u) = \frac{1}{u} \left(\lambda - \frac{b-1}{-\log u} \right). \quad (8.3)$$

Theorem 8.1 (Two complete monotone sectors). *Let $\eta_{\lambda,b}(\rho)$ denote the normalized angular zero for $F_b^{(\lambda)}$. If $-1 < \lambda \leq 0$ and $b \geq 1$, excluding $(\lambda, b) = (0, 1)$, it is strictly decreasing on $0 < \rho \leq 1$. If $\lambda \geq 0$ and $0 < b \leq 1$, excluding the same point, it is strictly increasing. At $(0, 1)$ it is constant $1/2$. In all these cases the analytic value at zero is*

$$\eta_{\lambda,b}(0) = \frac{1}{3} \left[1 + \left(\frac{1+\lambda}{2+\lambda} \right)^b \right]. \quad (8.4)$$

Proof. The gamma integral gives $\int_0^1 u^k w_{\lambda,b}(u) du = (k + \lambda + 1)^{-b}$, so (2.6) is exactly (8.1). Equation (8.3) gives strict decreasing or increasing behavior in the indicated sectors. Apply Theorem 5.2. Formula (4.7) gives (8.4). $\square$

The mixed sectors $\lambda > 0, b > 1$ and $-1 < \lambda < 0, 0 < b < 1$ are not covered by this monotone-curvature criterion. In those sectors the weight can change its direction of monotonicity; no radial sign is inferred merely from the sign of its first Taylor coefficient.

9 Exact sums over all inner orders

9.1 A weight-generating transport identity

Theorem 9.1 (Order generation is a Hurwitz shift). *For $|\tau| < 1$ and $z \in \mathbb{C} \setminus [1, \infty)$,*

$$\sum_{m=1}^{\infty} \tau^{m-1} F_m(z) = F_1^{(-\tau)}(z). \quad (9.1)$$

For a complex shift on the right, the function is defined by its disk series with denominators $k - \tau$ and its Green-integral continuation. The convergence in (9.1) is locally uniform jointly in z and τ .

Proof. In the disk, absolute convergence permits summation in the inner order:

$$\sum_{m \geq 1} \frac{\tau^{m-1}}{k^m} = \frac{1}{k - \tau}, \quad k \geq 1.$$

This proves the coefficient identity. The same transport is visible at the weight level:

$$\sum_{m \geq 1} \tau^{m-1} \frac{(-\log u)^{m-1}}{(m-1)!} = u^{-\tau}.$$

To justify continuation and local uniformity, let $K \subset \mathbb{C} \setminus [1, \infty)$ be compact and set

$$\delta_K = \min_{z \in K, 0 \leq u \leq 1} |1 - zu| > 0, \quad C_K = \frac{\max_{z \in K} |z|^2}{2\delta_K^2}.$$

Each Q_m has mass $1/2$, so (2.5) gives $|F_m(z)| \leq C_K$, uniformly in the positive integer m . The Weierstrass test proves the required convergence for $|\tau| \leq q < 1$. The complex Green weight $u^{-\tau}$ is integrable since $\operatorname{Re} \tau < 1$, and its integral is analytic in these parameters. The identity theorem extends the disk equality to the slit plane. $\square$

The same proof gives an explicit truncation bound:

$$\left| \sum_{m>M} \tau^{m-1} F_m(z) \right| \leq C_K \frac{|\tau|^M}{1-|\tau|}. \quad (9.2)$$

Thus the all-order identity has a simple independently reproducible analytic error contract.

9.2 Half shifts and parity-separated identities

Let

$$A(z) = \operatorname{atanh} \sqrt{z}.$$

The individual square root need not be chosen globally in the slit plane. In every formula below, only $A(z)^2$ and $\sqrt{z} A(z)$ appear. These are the single-valued analytic continuations from zero of their power series, so the formulas are unambiguous at negative real arguments as well.

Proposition 9.2 (Two elementary shifted functions). *On $\mathbb{C} \setminus [1, \infty)$,*

$$F_1^{(-1/2)}(z) = 2A(z)^2 - 4\sqrt{z} A(z) - 2\log(1-z), \quad (9.3)$$

$$F_1^{(1/2)}(z) = 2A(z)^2 + 2\log(1-z). \quad (9.4)$$

Proof. Differentiating (8.1) inside the disk gives

$$\frac{d}{dz} F_1^{(\lambda)}(z) = \frac{1}{1-z} \sum_{k \geq 1} \frac{z^k}{k + \lambda}. \quad (9.5)$$

The two inner sums at $\lambda = -1/2$ and $1/2$ are respectively $2\sqrt{z} A(z)$ and $2(A(z)/\sqrt{z} - 1)$. With $z = r^2$, differentiation of the proposed expressions gives $4r^2 \operatorname{atanh} r / (1 - r^2)$ and $4(\operatorname{atanh} r - r) / (1 - r^2)$, which agree with (9.5) after the chain rule. Both expressions vanish at zero. Equality follows in the disk and then by analytic continuation. $\square$

Theorem 9.3 (All odd and even inner orders). *On $\mathbb{C} \setminus [1, \infty)$,*

$$\sum_{j=0}^{\infty} 4^{-j} F_{2j+1}(z) = 2A(z)^2 - 2\sqrt{z} A(z), \quad (9.6)$$

$$\sum_{j=0}^{\infty} 4^{-j} F_{2j+2}(z) = -4\sqrt{z} A(z) - 4\log(1-z). \quad (9.7)$$

After the first J terms ($j = 0, \dots, J-1$), either tail has absolute value at most $(4C_K/3)4^{-J}$ for $z \in K$.

Proof. Take the half-sum and the difference, respectively, of (9.1) at $\tau = 1/2$ and $\tau = -1/2$. Insert (9.3)–(9.4). The tail estimate follows by the geometric series and $|F_m(z)| \leq C_K$. $\square$

Corollary 9.4 (Four constant identities). *In addition to (1.6)–(1.7),*

$$\sum_{j \geq 0} 4^{-j} F_{2j+1}(1/4) = \frac{\log^2 3 - \log 3}{2}, \quad (9.8)$$

$$\sum_{j \geq 0} 4^{-j} F_{2j+2}(1/4) = 8 \log 2 - 5 \log 3. \quad (9.9)$$

At $z = -1$, either parity tail is at most $(2/3)4^{-J}$; at $z = 1/4$ it is at most $(2/27)4^{-J}$.

Proof. At $z = -1$, take $\sqrt{z} = i$, so $A(z) = i\pi/4$. At $z = 1/4$, $\sqrt{z} = 1/2$ and $A(z) = \log 3/2$. Substitution gives the identities. The constants in the tail bound are $C_{\{-1\}} = 1/2$ and $C_{\{1/4\}} = 1/18$. $\square$

These identities organize an infinite family of different weights into a closed function. They do not express each separate $F_m(-1)$ in an elementary basis, and hence do not settle finite-weight period independence or the S6/S8 questions.

9.3 Positive integer shifts reduce to ordinary polylogarithms

For completeness, positive integral shifts give another exact family that is useful for checking implementations. Put $L = \text{Li}_1(z) = -\log(1-z)$ and $P_j(z) = \sum_{k=1}^j z^k/k$.

Proposition 9.5 (Finite reduction for integral shifts). *For every integer $m \geq 1$,*

$$F_1^{(m)}(z) = \frac{1}{2}L^2 + \text{Li}_2(z) - \frac{L}{m} + \sum_{j=1}^{m-1} \frac{z^{-j}}{j} (P_j(z) - L). \quad (9.10)$$

Every apparent pole at zero in the finite sum is removable.

Proof. Since $P_j - L = -\sum_{k \geq j+1} z^k/k$, each summand is analytic at zero. The coefficient of z^n , $n \geq 1$, on the right is

$$\frac{H_{n-1}}{n} + \frac{1}{n^2} - \frac{1}{mn} - \sum_{j=1}^{m-1} \frac{1}{j(n+j)} = \frac{H_{n+m-1} - H_m}{n}.$$

Use $1/(j(n+j)) = (1/j - 1/(n+j))/n$ for the simplification. This is exactly the coefficient in (8.1); the constant terms vanish. Analytic continuation proves the formula on the slit plane. $\square$

10 Verification, rational certificates, and numerical illustration

10.1 What is proved by which evidence

All continuum statements above are proved by the analytic arguments. The computational package has three different roles. Exact finite coefficient and moment tests catch transcription errors. Rational intervals with an infinite-series tail bound certify selected numerical roots. Floating-point quadrature illustrates the geometry and checks against a different evaluation formula; it is not an interval proof. The three kinds of evidence are reported separately.

The executed standard-library verifier produced 180 exact odd-moment sign checks (integer $b = 1, \dots, 12$, odd orders $3, 5, \dots, 31$), 768 coefficient equalities for the half-shift identities, 960 coefficient equalities for positive integer shifts, and 30 checks of $K(1, b) = -2M_3$. Their total is 1,938. A separate SymPy script checks five symbolic identities, including the universal fourth-order radial coefficient in five independent raw moments. These are finite replays, not proof-assistant formalizations.

10.2 A rational certificate for selected angular zeros

Fix rational $0 < \rho < 1$ and rational $0 < \eta < 1$. Put $x = \rho^2 \eta$ and $s = \rho^2$. Although the imaginary coordinate of $z = x + i\sqrt{s - x^2}$ need not be rational, the normalized powers

$$P_0 = 0, \quad P_1 = 1, \quad P_n = 2xP_{n-1} - sP_{n-2}$$

are rational, and equal $\text{Im}(z^n)/\text{Im } z$. Thus

$$J = \frac{\text{Im } F_b(z)}{\text{Im } z} = \sum_{n \geq 2} \frac{H_{n-1}^{(b)}}{n} P_n. \quad (10.1)$$

The elementary inequality $|\sin(n\theta)| \leq n \sin \theta$ for $0 < \theta < \pi$ gives $|P_n| \leq n\rho^{n-1}$. Since $H_{n-1}^{(b)} \leq n-1$ for all $b > 0$, the tail after outer index N is bounded by the rational number

$$B_N(\rho) = \rho^N \left(\frac{N}{1-\rho} + \frac{\rho}{(1-\rho)^2} \right). \quad (10.2)$$

For positive integral b , every retained coefficient is rational. For $b = 1/2$ and $1/4$, the verifier encloses each k^{-b} by rational numbers with denominator $D = 10^{60}$, using integer square roots only. More generally, if $b = p/q$ and q is a power of two, it finds a with

$$a^q k^p \leq D^q < (a+1)^q k^p,$$

which proves $a/D \leq k^{-b} < (a+1)/D$. Harmonic partial sums are then propagated as rational intervals, with multiplication signs handled exactly. No floating-point approximation is used to choose a sign in the verifier.

For each proposed pair $\eta_- < \eta_+$, the script proves that the upper enclosure of the truncated series at η_- plus B_N is negative, and that the lower enclosure at η_+ minus B_N is positive. Lemma 5.1 then places the unique root strictly between them. The executed suite certifies 18 brackets at

$$b \in \{1/4, 1/2, 2, 3, 4, 8\}, \quad \rho \in \{1/4, 1/2, 3/4\}, \quad N = 192.$$

Every bracket has width $3 \cdot 10^{-10}$. All endpoints, exact sign margins, and rational tail bounds are in the JSON receipt. For example,

$$\begin{aligned} \frac{829818637}{2000000000} &< \eta_2(1/2) < \frac{1037273297}{2500000000}, \\ 0.4149093185 &< \eta_2(1/2) < 0.4149093188. \end{aligned} \quad (10.3)$$

The decimal line is just the terminating decimal form of the rational endpoints; it does not introduce a separate rounding assertion.

10.3 Independent floating-point diagnostics

For numerical illustration we use the single-integral identity

$$F_b(z) = \int_0^1 w_b(t) \frac{\log(1-tz)/t - \log(1-z)}{1-t} dt. \quad (10.4)$$

b	$\eta_b(0)$	$\eta_b(1/4)$	$\eta_b(1/2)$	$\eta_b(3/4)$	$\eta_b(1)$
1/4	0.6136321384	0.6142725544	0.6163248312	0.6202820025	0.6275145907
1/2	0.5690355937	0.5694172146	0.5706248931	0.5728838669	0.5767481557
1	0.5000000000	0.5000000000	0.5000000000	0.5000000000	0.5000000000
2	0.4166666667	0.4162313162	0.4149093186	0.4126523072	0.4093780509
4	0.3541666667	0.3533689224	0.3509838169	0.3470392575	0.3415922469
8	0.3346354167	0.3337176588	0.3309859468	0.3265081195	0.3204054688

Table 1: Illustrative quadrature values. The separate rational receipts certify selected interior-radius brackets, not every printed digit of every entry. At $b = 1$ the value is exactly $1/2$.

It follows by integrating $F'_b(z) = \text{Li}_b(z)/(1-z)$ from zero and inserting the gamma integral for Li_b . Equivalently, one may expand the two logarithms and compare the coefficients directly. The apparent $t = 0$ and $t = 1$ singularities of the kernel are removable for fixed slit-plane z .

With $t = e^{-T}$, generalized Gauss–Laguerre quadrature applies to the weight $e^{-T}T^{b-1}/\Gamma(b)$. The diagnostic script uses 256 nodes, compares with 128 nodes, and also compares 24 values against an independent 70-digit, 1,024-term direct double-polylogarithm series strictly inside the disk. The maximum observed absolute discrepancy against that series was $3.38 \cdot 10^{-12}$. The script checks the predicted monotonic direction on a 606-point grid, including the analytic $\rho = 0$ values. This finite grid has no role in proving Theorem 1.1.

10.4 Reproduction and artifact contract

The package is additive and does not change the repository remotely. From its root, use fresh output directories:

```
python verification/verify_exact.py --output-dir /tmp/polylog-exact
python verification/verify_symbolic.py --output-dir /tmp/polylog-symbolic
python verification/verify_numeric.py --output-dir /tmp/polylog-numeric \
  --figures-dir /tmp/polylog-figures
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

The exact verifier uses only the Python standard library. The symbolic script uses SymPy; numerical diagnostics use NumPy, SciPy, mpmath, and Matplotlib. Each script refuses to overwrite its own named result file, so replaying into a new directory preserves received evidence. The included receipts record what was actually executed in this session. No checksum-normalization workflow is required.

11 Audit findings and integration guidance

11.1 A status correction, not a correction to a valid formula

In the inspected local-radius section, the whole-radius $a = 1$ branch is still framed as a conjectural extension of the local result. After this article is reviewed, the appropriate update is narrow: replace that branch’s open status by Theorem 1.1, and retain the distinction between it and all larger outer orders. The existing local coefficient agrees with (7.3); its proof is not invalidated. The earlier Bernstein calculation remains useful evidence and historical context.

The section on Cartesian motion proves an outward abscissa law and explicitly distinguishes it from normalized-radius motion [4]. That distinction is correct. The present proof fills one part of the remaining normalized question by (5.7); it does not reinterpret the Cartesian theorem as something stronger.

11.2 Two small notation repairs in the transport proof

In the inspected `05-real-positive-kernel.tex`, the proof of the Hurwitz–polylogarithm transport theorem says to use the Hurwitz formula “with $q = n$ ”, although its argument is named x ; this should read “with $x = n$ ”. The same proof invokes “boundedness of h ”, although the regularized kernel in that argument is $q(t)$; it should read “boundedness of q ” [5]. These are local notation repairs, not refutations of the transport identity. They are not asserted to be first discoveries and may already be tracked in incoming work. The integration notes keep them separate from the new mathematical claim.

11.3 Why the same proof does not automatically cover larger outer orders

For integer $a \geq 2$, the source signed kernels satisfy $\kappa'_{a,b}(u) = -\kappa_{a-1,b}(u)/u$. If $Q_{a,b}(u) = \int_u^1 \kappa_{a,b}(v) dv$, then

$$Q''_{a,b}(u) = \frac{\kappa_{a-1,b}(u)}{u}. \quad (11.1)$$

The right side changes sign. Thus $Q_{a,b}$ is not generally concave, and Lemma 3.1 cannot simply be invoked with that density. This identifies the hypothesis that fails in a tempting extension; it is not a claim that the higher-index conjecture is false or inaccessible. Other representations or a more flexible sign criterion may still prove it.

11.4 What this audit does not certify

The audit is local to the named analytic material and the displayed formulas. It does not certify the entire 379-page manuscript, all historical reports, all incoming ZIP members, period independence, or the correctness of unrelated analytic continuations. The article has not been independently refereed. All source-linked assertions can be checked against the recorded paths and pin; the new theorems have complete proofs here rather than relying on numerical plausibility.

12 Further research questions

1. **Higher outer orders.** Find a replacement for strict concavity in the proof of Lemma 3.1, compatible with the single change of curvature in (11.1). A useful intermediate target is a signed reflected-difference criterion that still controls the implicit radial denominator and numerator separately. Any progress should be compared with the latest incoming large-order and all-depth results before assigning an open range.
2. **The subunit-inner-order curve outside the disk.** For $0 < b < 1$, determine the sign changes of $m_b(y)m'_b(y) + y$, which controls $R'_b(y)$ in Corollary 5.3. Does each circle meet the upper imaginary-zero curve at most once, or can the inverse-mode graph produce radial turning? The complete parametrization (5.8) reduces the problem to one scalar analytic function, but its sign is not settled here.

3. **Sharp quantitative motion bounds.** Formula (5.7) proves strictness but does not give a simple optimal lower bound for $|\eta'_b(\rho)|$. Bound the reflected crossing v_0 and the two sign-definite integrals to obtain uniform estimates on compact (b, ρ) ranges, especially as $b \rightarrow 1$ or $\rho \rightarrow 1$.
4. **Complete curvature classification.** The exact fourth-order coefficient $3M_5 - (12M_2 + 2\mu_b^2)M_3$ is negative at $b = 2$ and positive at $b = 4$. Determine all its zeros for $b > 1$, with rigorous enclosures, and study whether they govern changes in the full radial curvature rather than only near the origin. This question must not be confused with the already proved first-derivative sign.
5. **The mixed shifted sectors.** Classify normalized-radius motion for $\lambda > 0, b > 1$ and $-1 < \lambda < 0, 0 < b < 1$. In these regions (8.3) has a zero. A natural task is to classify the number of reflected sign changes and the possible number of radial turning points before attempting a global phase diagram.
6. **More general smoothing kernels.** The proof of (4.6) uses that the logarithmic scale derivative of the positive stationary weight is increasing in the reflection distance. Formulate a maximal kernel class with that property. Gaussian kernels are a natural comparison; kernels whose scale score changes direction may expose necessary rather than merely sufficient shape hypotheses.
7. **Rational shifts and finite-weight consequences.** The order-generating identity (9.1) converts rational values of τ to rational shifts. Determine which such shifts have useful logarithmic or cyclotomic descriptions and whether their coefficient extractions yield new finite-weight identities. Infinite weight sums alone do not prove the S6 or S8 reductions.
8. **Formalization and certified continuation.** The finite rational root verifier is already suitable for independent reimplementations. Formalize the Green regularity lemma, the signed zero-mass comparison, the reflected single crossing, and the implicit radial derivative as separate components. Extend the rational certificates toward $\rho = 1$ using a rigorously accelerated series or positive-kernel quadrature, rather than labeling ordinary quadrature as certified.

13 Conclusion

The leading-index-one normalized-radius conjecture is resolved by a shape theorem for a positive Green density. The decisive information is not simply positivity or uniqueness of an angular zero: it is the ordered sign change of a reflected density, followed by a careful change from inverse-height differentiation to radial differentiation. This proves all three inner-order regimes on the full disk, gives a larger global domain when $b \geq 1$, and supplies both moment inequalities and exact all-order identities. The proof also exposes a real limit of stronger conjectures through opposite local curvature signs. The remaining higher-index, mixed-shift, and finite-weight arithmetic questions are separated from these proved statements rather than being hidden behind a numerical residual.

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